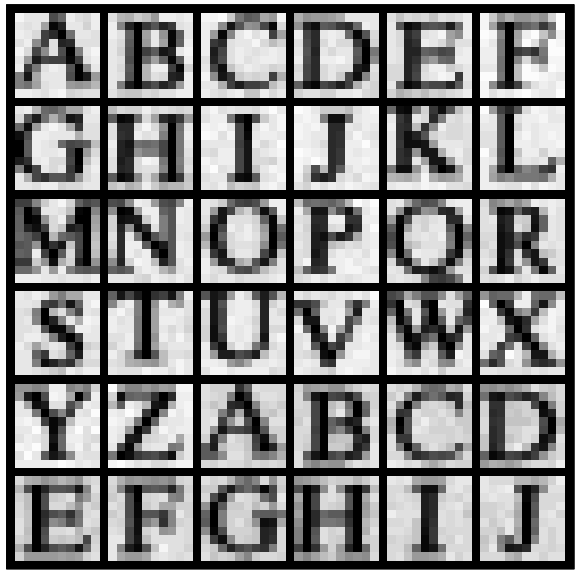

FIGURE 26-8
Example image of text. Identifying letters in images of text is one of the classic pattern recognition problems. In this example, each letter is contained in a 10×10 pixel image, with 256 gray levels per pixel. The database used to train and test the example neural network consists of 50 sets of the 26 capital letters, for a total of 1300 images. The images shown here are a portion of this database.

of each 10×10 image in the database could indicate what the letter is. For example, images 0 to 49 might all be an "A", images 50-99 might all be a "B", etc.

For this demonstration, the neural network will be designed for an arbitrary task: determine which of the 10×10 images contains a *vowel*, i.e., A, E, I, O, or U. This may not have any practical application, but it does illustrate the ability of the neural network to learn very abstract pattern recognition problems. By including ten examples of each letter in the training set, the network will (hopefully) learn the key features that distinguish the target from the nontarget images.

The neural network used in this example is the traditional three-layer, fully interconnected architecture, as shown in Figs. 26-5 and 26-6. There are 101 nodes in the input layer (100 pixel values plus a bias node), 10 nodes in the hidden layer, and 1 node in the output layer. When a 100 pixel image is applied to the input of the network, we want the output value to be close to *one* if a vowel is present, and near *zero* if a vowel is not present. Don't be worried that the input signal was acquired as a two-dimensional array (10×10), while the input to the neural network is a one-dimensional array. This is *your* understanding of how the pixel values are interrelated; the *neural network* will find relationships of its own.

Table 26-2 shows the main program for calculating the neural network weights, with Table 26-3 containing three subroutines called from the main program. The array elements: X1[1] through X1[100], hold the input layer values. In addition, X1[101] always holds a value of 1, providing the input to the bias node. The output values from the hidden nodes are contained